WAGENINGEN UNIVERSITY & RESEARCH

Postdoc on computational (meta)genomic analysis



ABOUT US

Career at WUR

- Plenty of variety and challenges
- Pleasant working atmosphere
- Time for personal development
- Proper work-life balance
- Internationally leading

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natural diversity that enables plants and their associated microbiomes to adapt to environmental change. Ancient environmental DNA (eDNA) provides a unique record of past ecosystems, preserving genetic traces of microorganisms and plants that interacted with crops and their wild relatives over thousands of years. In this project, you will develop and apply computational omics approaches to uncover the molecular mechanisms underlying ancient plant–microbiome interactions, with a particular focus on biosynthetic gene clusters and other functional genes involved in chemical communication, defence, and adaptation. You will analyse large-scale ancient metagenomic datasets to identify and characterize microbial and plant-derived biosynthetic pathways, predict their ecological functions, and reconstruct the (co-)evolutionary dynamics of traits such as microbiome recruitment, metabolite-mediated interactions, pathogenicity and disease resistance. This will involve using and further adapting state-of-the-art genome mining approaches, including biosynthetic gene cluster identification, gene cluster family classification, and functional prediction, to work with highly fragmented and complex eDNA datasets.

This project is part of the larger European AEGIS consortium, bringing together expertise in ancient DNA, plant science, microbiome research, computational biology, and crop improvement. You will work closely with researchers across the consortium to translate computational predictions into biological insights and identify ancient genetic traits that may contribute to future climate-resilient agriculture. Within Wageningen University, you will be embedded in the Bioinformatics chair group and collaborate with researchers developing innovative approaches for microbiome and genome mining. The research is expected to deliver fundamental insights into the evolutionary history of plant–microbiome interactions, novel computational strategies for analysing ancient omics data, and candidate genes and pathways that can ultimately support the development of more resilient crops.

Your duties and responsibilities include:

- computational (meta)genome mining and quantitative data analysis of biosynthetic and catabolic genes and clusters, pathogenicity genes and other functional genes relevant for host-microbe interactions across ancient DNA samples;
- biological interpretation of the data to elucidate evolution of plant-microbiome interactions and metabolites involved therein;
- development or improvement of existing or new algorithms or computational strategies;
- collaboration with international partners.

Your team

You will work within the Bioinformatics chair group at Wageningen University, in the team of Marnix Medema. The expertise of the Medema research group within the Bioinformatics chair group lies in developing and applying computational approaches to identify metabolic pathways and connect them to specific molecules and phenotypes.

You will work here

Wageningen UR has a global reputation for its ground-breaking research and innovative education in the fields of food and health, sustainable agrosystems, the rural environment, and societal development - all of which make an essential contribution to the quality of life, i.e. the supply of safe and healthy food and drink, on the one hand, and the chance to live, work and play in a balanced ecosystem with a large variety of plants and animals on the other. Within Wageningen UR, the Plant Sciences Group (PSG) brings together science education and scientific, strategic and applied research in the fields of biology, of plants in relation to their environment, of plant-related organisms and of plant production. This project will be carried out at the Bioinformatics Group within PSG. This research group focuses on fundamental and applied bioinformatics research in the green life sciences. In particular, they develop and apply novel computational methods for the analysis and integration of –omics data. The group has a strong track record in (integrative) computational omics analysis, algorithm development, machine learning and scientific data infrastructure. There are many national and international collaborations with researchers studying plant and microbe metabolism, biotechnology and genomics.

- a strong academic record (PhD) in bioinformatics or a related subject;
- affinity with the study of plants/microbiomes and/or metabolism and analysis of omics data;
- proven proficiency in programming;
- strong statistical skills and;
- excellent oral and written communication skills in English;

In our international working environment there is an increasing amount of communication in English. For this position it is about **language level** C1.

We offer

Wageningen University & Research offers excellent **terms of employment**. A few highlights from our Collective Labour Agreement include:

- sabbatical leave, study leave, and partially paid parental leave;
- working hours that can be discussed and arranged so that they allow for the best possible work-life balance;
- there is a strong focus on vitality and you can make use of the sports facilities available on campus for a small fee;
- a fixed year-end bonus of 8.3%;
- excellent pension scheme.

In addition to these first-rate employee benefits, you will of course receive a good salary. Depending on your experience, we offer a competitive gross salary of between € 3,706.- and € 5,760.- for a full-time working week of 38 hours, in accordance with the **Collective Labour Agreements for Dutch Universities (CAO-NU)** (scale 10). As for your contract, we initially offer a one-year contract that can be extended - if possible within (company) organizational circumstances - on good performance, for the duration of the project (3 years in total).

We encourage development and internal mobility within our organisation. Our recruitment and selection policy sets out the conditions that apply specifically to you as a (former) employee. If you have any questions, we are happy to help.

You will work on the greenest and most innovative campus in the Netherlands, in an international and open working environment.

Coming from abroad

Wageningen University & Research is the university and research centre for life sciences. The themes we deal with are relevant to everyone around the world and Wageningen, therefore, has a large international community and a lot to offer to international employees.

Because we expect you to work and live in the Netherlands our team of advisors on Dutch immigration procedures will help you with the visa application procedures for yourself and, if applicable, for your family.

Feeling welcome also has everything to do with being well informed. Wageningen University & Research's **International Community** page contains practical information about what we can do to support international employees coming to Wageningen. Furthermore, our **Welcome Center** can assist you with any additional advice and information about for example housing, opening a bank account, dual career programs or schooling. Finally, certain categories of international staff may be eligible for a tax exemption on a part of their salary during the first five years in the Netherlands.

email: **marnix.medema@wur.nl**.
Questions about the procedure? Get in touch with Noorien Abbas, Corporate Recruiter, via email: **vacaturemeldingen.psg@wur.nl**.

Ready to apply?
Click on the application button next to the vacancy on our website. Only applications submitted through our website will be considered.

You can apply **up to and including MondayAugust 31, 2026**. The first interviews are scheduled for later 2026. Internal candidates are preferred to apply before August 4, 2026.

To apply, please send the following documents:

- Complete and up-to-date curriculum vitae;
- Motivation letter.

Additional files such as grades and transcripts are not required during this stage and will not be considered.

Procedure
As part of our selection process, an assessment may be incorporated within the procedure.

Welcome, safe, and valued
Wageningen University & Research (WUR) highly values diversity and inclusion because we believe that different insights lead to innovative solutions. We create a work environment where everyone feels welcome, safe, and appreciated, regardless of background, identity, or experience. Together, we are building a culture where everyone's unique contribution adds to the success of our organization.

We are

The mission of Wageningen University & Research is “To explore the potential of nature to improve the quality of life”. Under the banner Wageningen University & Research, Wageningen University and the specialised research institutes of the Wageningen Research Foundation have joined forces in contributing to finding solutions to important questions in the domain of healthy food and living environment. With its roughly 30 branches, 7,600 employees (6,700 fte) and 13,100 students and over 150,000 participants to WUR’s Life Long Learning, Wageningen University & Research is one of the leading organisations in its domain. The unique Wageningen approach lies in its integrated approach to issues and the collaboration between different disciplines.

Read the **5 reasons** why your future colleagues enjoy working at WUR and watch the video below to get an idea of our green campus!

We will recruit for the vacancy ourselves, so no employment agencies please. However, sharing in your network is appreciated.

WAGENINGEN UNIVERSITY & RESEARCH

PROFESSIONAL FIELD

Research & Science (incl. postdoc)

EDUCATIONAL LEVEL

Doctorate

APPLY BEFORE

1-9-2026

SALARY RANGE

€3706 - €5760

HOURS

38 - 38

Apply for this job